

Hypothesis Testing

Questions

Question 1: What is a null hypothesis (H_0) and why is it important in hypothesis testing?

A null hypothesis, denoted as H_0 , is a foundational concept in hypothesis testing. It represents the assumption that **there is no significant effect, difference, or relationship** between variables in a population — essentially, that any observed variation in a sample is due to random chance.

H_0 : The statement of “no change” or “no difference.” Example: If you’re testing whether a new teaching method improves student scores.

H_0 : The new method does *not* improve scores compared to the old one.

Importance in Hypothesis Testing

Starting Point for Analysis: It provides a baseline assumption that researchers attempt to challenge with data. The null hypothesis provides a clear, testable statement that forms the basis of all statistical testing.

Ensures Objectivity and Scientific Rigor By starting with the assumption that *no effect exists*, researchers avoid bias. This neutrality ensures that conclusions are drawn purely from evidence rather than expectations or personal beliefs.

Statistical Decision Framework: Hypothesis testing revolves around determining whether sample evidence is strong enough to reject H_0 in favor of the **alternative hypothesis (H_1)**.

Error Control: It helps quantify the risk of making incorrect decisions (Type I and Type II errors); **Type I error (α):** Rejecting H_0 when it’s actually true (false positive).

Type II error (β): Failing to reject H_0 when it’s false (false negative).

These error rates are essential for maintaining reliability and credibility in research.

Quantifies Evidence Strength The null hypothesis allows the use of **p-values** and **confidence intervals** to measure how strongly data contradict the assumption of no effect. This quantification transforms intuition into measurable evidence.

Supports Comparability Across Studies Because H_0 is standardized (“no effect” or “no difference”), results from different studies can be compared or replicated easily, a cornerstone of scientific progress.

The null hypothesis acts as the **anchor of scientific testing**.

Question 2: What does the significance level (α) represent in hypothesis testing?

The **significance level (α)** in hypothesis testing represents the **threshold for deciding whether to reject the null hypothesis (H_0)**. It quantifies the probability of making a **Type I error** — that is, rejecting H_0 when it is actually true.

α (alpha) is the **maximum acceptable probability of error** you're willing to tolerate when concluding that an effect exists. Common choices are **0.05**, **0.01**, or **0.10**, depending on how strict the test needs to be.

α Represents;

Risk Tolerance for False Positives

It defines how much risk you accept of wrongly claiming a significant result. Example: $\alpha = 0.05$ means you accept a 5 % chance of rejecting H_0 when it's true.

Decision Boundary It sets the **cut-off point** for the *p-value*.

- If $p \leq \alpha$, you **reject H_0** (evidence is statistically significant).
- If $p > \alpha$, you **fail to reject H_0** (evidence is insufficient).

Controls Confidence Level

The confidence level is $1 - \alpha$. So, $\alpha = 0.05$ corresponds to a **95 % confidence level**, meaning you're 95 % confident your conclusion isn't due to random chance.

Balances Sensitivity and Reliability- A smaller α (like 0.01) reduces false positives but increases the chance of missing real effects (Type II errors). A larger α (like 0.10) makes detection easier but less reliable.

Standardization Across Studies- Using a consistent α allows researchers to compare results and maintain uniform standards of statistical significance.

Question 3: Differentiate between Type I and Type II errors.

In hypothesis testing, decisions are made based on sample data — but there's always a chance of making a wrong decision. These wrong decisions are classified as **Type I** and **Type II errors**.

Type I Error (α Error)

- Occurs when the **null hypothesis (H_0) is true**, but we **reject it**.
- It's a **false positive** — we conclude there's an effect or difference when none actually exists.
- The probability of committing a Type I error is denoted by **α** , which is the **significance level** (commonly 0.05 or 5%).
- **Example:** A medical test wrongly indicates a patient has a disease when they are actually healthy.
- **Impact:** Leads to incorrect claims of significance; researchers may waste resources pursuing false leads.

Type II Error (β Error)

- Occurs when the **null hypothesis (H_0) is false**, but we **fail to reject it**.
- It's a **false negative** — we miss detecting a real effect or difference.
- The probability of committing a Type II error is denoted by β .
- The **power of a test** is $1-\beta$, representing the probability of correctly rejecting a false H_0 .
- **Example:** A medical test fails to detect a disease that the patient actually has.
- **Impact:** Real effects or relationships go unnoticed, leading to missed opportunities or incorrect conclusions.

Relationship Between α and β

- There's a trade-off between the two:
 - Lowering α (making the test more stringent) reduces false positives but increases the chance of false negatives (β).
 - Increasing α makes detection easier but less reliable.
- Researchers aim to balance both by choosing an appropriate **significance level** and **sample size** to minimize overall error risk.

Type I Error - Rejecting a true H_0 (false alarm). **Type II Error - Accepting a false H_0 (missed detection).**

Question 4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.

In hypothesis testing, the choice between a one-tailed and a two-tailed test depends on the **direction** of the research hypothesis.

1- One-Tailed Test

- **Meaning:** Tests for an effect in **only one direction** — either greater than or less than a specified value.
- **Hypothesis Form:**
 - $H_0: \mu = \mu_0$
 - $H_1: \mu > \mu_0$ (*right-tailed*) or $H_1: \mu < \mu_0$ (*left-tailed*)
- **Critical Region:** Entire significance level (α) lies in **one tail** of the distribution.
- **When Used:** When prior theory or evidence suggests the effect can occur **only in one direction**.

Example:

A company claims its new training program **increases** employee productivity.

- $H_0: \mu = 50$ units/hour
- $H_1: \mu > 50$ units/hour - **Right-tailed test**

Two-Tailed Test

- Meaning: Tests for an effect in both directions, whether the sample mean is either greater or less than the hypothesized value.
- Hypothesis Form:
 - $H_0: \mu = \mu_0$
 - $H_1: \mu \neq \mu_0$
- Critical Region: Significance level (α) is split equally between both tails ($\alpha/2$ in each).
- When Used: When the researcher wants to check for any difference, regardless of direction.
- Example:

A manufacturer wants to test if a new machine produces parts with a different average weight than the old one.

- $H_0: \mu = 100$ g
- $H_1: \mu \neq 100$ g - Two-tailed test

One-tailed test - directional (tests “greater than” or “less than”).

Two-tailed test- non-directional (tests “different from”).

Question 5: A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim.

A company claims that the **average time to resolve a customer complaint is 10 minutes**.

A random sample of **9 complaints** gives:

- Sample mean ($\bar{x}$) = 12 minutes
- Sample standard deviation (s) = 3 minutes
- Significance level (α) = 0.05

We need to **test the company's claim**.

State the Hypotheses

$H_0: \mu = 10$ (The average resolution time is 10 minutes)

$H_1: \mu \neq 10$ (The average resolution time is not 10 minutes)

This is a **two-tailed test** because we are checking for any difference (higher or low)

Test Statistic

Since the sample size is small ($n < 30$) and population standard deviation is unknown, we use the **t-test**.

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

Substitute the values:

$$t = 12 - 10 / 3\sqrt{9}$$

$$= 2/1$$

$$= 2.0$$

Degrees of Freedom

$$df = n - 1$$

$$= 9 - 1 = 8$$

Critical Value;

At $\alpha = 0.05$ for a **two-tailed test** and $df = 8$:

$$t_{\text{critical}} = \pm 2.306$$

Decision Rule;

- If $|t_{\text{calculated}}| > t_{\text{critical}}$, **reject** H_0 .
- Otherwise, **fail to reject** H_0 .

Here:

$$|t_{\text{calculated}}| = 2.0 < 2.306$$

Since the calculated t-value does **not exceed** the critical value, we **fail to reject** H_0 .

Interpretation- At the 5 % significance level, there is **no sufficient evidence** to conclude that the average time to resolve a customer complaint differs from 10 minutes.

Question 6: When should you use a Z-test instead of a t-test?

In hypothesis testing, both **Z-tests** and **T-tests** are used to compare sample statistics with population parameters. The choice between them depends mainly on **sample size** and **knowledge of the population standard deviation (σ)**.

Z-Test

Used when:

- The **population standard deviation (σ)** is **known**.
- The **sample size is large** ($n \geq 30$).
- The data follows a **normal distribution** or the sample is large enough for the Central Limit Theorem to apply.

Example: Suppose a company knows that the population standard deviation of delivery times is 5 minutes. It wants to test whether the mean delivery time differs from 30 minutes, Use a **Z-test** because σ is known and n is large.

T-Test

Used when:

- The **population standard deviation (σ)** is **unknown**.
- The **sample size is small** ($n < 30$).
- The data is approximately **normally distributed**.

Example: A researcher collects data from 20 students to test if their average study time differs from 3 hours, but the population standard deviation is unknown, Use a **T-test** because σ is unknown and n is small.

Key differences;

Aspect	Z-Test	T-Test
Population Standard Deviation	Known (σ)	Unknown (use sample s)
Sample Size	Large ($n \geq 30$)	Small ($n < 30$)

Distribution	Normal (Z)	Student's t
Variability	Less variable	More variable (accounts for sample uncertainty)
Example	Testing mean delivery time with known σ	Testing mean study time with unknown σ

Use a **Z-test** when σ is known and the sample is large; use a **T-test** when σ is unknown or the sample is small. Both tests help determine whether the sample mean significantly differs from the hypothesized population mean.

Question 7: The productivity of 6 employees was measured before and after a training program.

employee	before	after
1	50	55
2	60	65
3	58	59
4	55	58
5	62	63
6	56	59

At $\alpha = 0.05$, test if the training improved productivity.

Solution ;

Employee	Before	After	Difference (After – Before)
1	50	55	5

2	60	65	5
3	58	59	1
4	55	58	3
5	62	63	1
6	56	59	3

Stating the Hypotheses

$H_0: \mu_d = 0$ (No improvement in productivity)

$H_1: \mu_d > 0$ (Training improved productivity)

This is a **right-tailed paired t-test**.

Computing the Mean and Standard Deviation of Differences

$$\bar{d} = 5 + 5 + 1 + 3 + 1 + 3 / 6$$

$$= 18 / 6$$

$$= 3$$

Now computing the sample standard deviation of differences:

$$s_d = \sqrt{[\Sigma(d_i - \bar{d})^2 / (n - 1)]}$$

d_i	d_i - d̄	(d_i - d̄)²
5	2	4
5	2	4
1	-2	4

3	0	0
1	-2	4
3	0	0

$$\sum (d_i - \bar{d})^2 = 16$$

$$sd = \sqrt{(16/5)}$$

$$= \sqrt{3.2}$$

$$= 1.79$$

Calculating the Test Statistic

$$t = (\bar{d} - 0) / (s_d / \sqrt{n})$$

$$= 3 / 1.79 / \sqrt{6}$$

$$= 3 / 0.73$$

$$= 4.11$$

Degrees of Freedom

$$df = n - 1 = 6 - 1 = 5$$

Critical Value;

At $\alpha = 0.05$ for a **right-tailed test** with $df = 5$

$$t_{critical} = 2.015$$

Decision Rule

Since $t_{calculated} = 4.11 > t_{critical} = 2.015$,

We reject H_0 .

At the 5 % significance level, there is **sufficient evidence** to conclude that the **training program improved employee productivity**.

Question 8: A company wants to test if product preference is independent of gender.

gender	Product A	Product B	total
male	30	20	50
female	10	40	50
total	40	60	100

At $\alpha = 0.05$, test independence

Stating the Hypotheses;

H₀:Product preference is independent of gender.

H₁:Product preference is not independent of gender.

Computing Expected Frequencies;

Expected frequency for each cell:

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

Gende r	Product A	Product B
Male	$E = \frac{50 \times 40}{100} = 20$	$E = \frac{50 \times 60}{100} = 30$
Femal e	$E = \frac{50 \times 40}{100} = 20$	$E = \frac{50 \times 60}{100} = 30$

Computing the Chi-Square Statistic;

$$\chi^2 = \sum (O - E)^2 / E$$

Gende r	Product A	Product B	Calculation
Male	O=30,E=20	O=20,E=30	$\frac{(30-20)^2}{20} + \frac{(20-30)^2}{30}$

			20 30
Female	O=10,E=20	O=40,E=30	$\frac{(10-20)^2}{20} + \frac{(40-30)^2}{30}$

$$\chi^2 = \frac{100}{20} + \frac{100}{30} + \frac{100}{20} + \frac{100}{30}$$

$$\chi^2 = 5 + 3.33 + 5 + 3.33$$

$$= 16.66$$

Degrees of Freedom;

$$df = (r-1)(c-1) = (2-1)(2-1) = 1$$

Critical Value;

At $\alpha = 0.05$ and $df=1$:

$$\chi^2_{\text{critical}} = 3.841$$

Decision Rule;

Since $\chi^2_{\text{calculated}} = 16.66 > 3.841$,

We reject H_0 .

At the 5 % significance level, there is **sufficient evidence** to conclude that **product preference is not independent of gender**. In other words, **gender influences product preference**.